24-Camera Surveillance Operations Center

Complete Linux Server Deployment, Systemd / Systemctl Configuration, and VAPT Hardening Manual

1. SYSTEM SPECIFICATIONS & COMPATIBILITY

Engineered to run seamlessly across **ANY modern Linux distribution** (Ubuntu 20.04/22.04/24.04, Debian 11/12, RHEL 8/9, Rocky Linux, AlmaLinux, CentOS Stream, Fedora, openSUSE):

- **Architecture:** x86_64 (64-bit Linux).
- **Processor:** Minimum 4 CPU Cores (8 Cores recommended for high-volume 24-channel viewing).
- **System Memory:** Minimum 8 GB RAM (16 GB recommended).
- **Storage:** Fast NVMe/SSD with minimum 20 GB free space.
- **Runtime Stack:** Node.js LTS (v20+) + Embedded MediaMTX (v1.21.2) RTSP/WebRTC gateway.

2. NETWORK & FIREWALL PORT ALLOCATION

Port	Protocol	Scope	Service / Description
3000	TCP	Internal / Proxy	Node.js Express Web Dashboard & REST API
80 / 443	TCP	Public / LAN	HTTP / HTTPS Nginx Reverse Proxy & WebSockets
8554	TCP	LAN / Subnet	MediaMTX RTSP Ingestion & Camera Proxy
8889	TCP	LAN / WAN	MediaMTX WebRTC (WHEP) HTTP Signaling
8189	UDP	LAN / WAN	MediaMTX WebRTC ICE Media Streaming
8888	TCP	LAN / WAN	MediaMTX Low-Latency HLS Fallback Stream
9997	TCP	**Localhost Only**	MediaMTX Dynamic REST Control API (Bound to 127.0.0.1)

3. AUTOMATED 1-COMMAND PRODUCTION SETUP

To deploy the entire stack automatically on any Linux server (creates unprivileged user, extracts files, installs dependencies, and configures systemd):

```
# 1. Unzip the deployment archive
unzip inncctv-enterprise-linux.zip -d /tmp/inncctv-pkg
cd /tmp/inncctv-pkg

# 2. Run the automated installer with root privileges
sudo bash deploy/setup-server.sh
```

4. STEP-BY-STEP MANUAL LINUX DEPLOYMENT

If you prefer full control over every step on your Linux server, run the following commands:

Step 4.1: Install Prerequisites (Ubuntu/Debian)

```
sudo apt-get update -y && sudo apt-get upgrade -y
sudo apt-get install -y curl wget unzip ffmpeg ufw nginx
```

Step 4.1 (Alternative): Install Prerequisites (RHEL / Rocky / AlmaLinux)

```
sudo dnf check-update -y && sudo dnf install -y epel-release
sudo dnf install -y curl wget unzip ffmpeg firewalld nginx
```

Step 4.2: Install Node.js LTS (v20+)

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt-get install -y nodejs   # On RHEL: sudo dnf install -y nodejs
node -v && npm -v
```

Step 4.3: Create Dedicated System User (Security Hardening)

```
sudo useradd -r -s /usr/sbin/nologin -d /opt/inncctv inncctv
```

Step 4.4: Deploy Files to `/opt/inncctv`

```
sudo mkdir -p /opt/inncctv
sudo unzip inncctv-enterprise-linux.zip -d /opt/inncctv/
```

Step 4.5: Install Node.js Production Dependencies

```
cd /opt/inncctv
sudo npm install --omit=dev
```

Step 4.6: Set Executable Flags & Folder Permissions

```
sudo chmod +x /opt/inncctv/bin/mediamtx
sudo chmod +x /opt/inncctv/start.sh /opt/inncctv/stop.sh
sudo chown -R inncctv:inncctv /opt/inncctv
```

5. SYSTEMCTL CONFIGURATION & SERVICE MANAGEMENT (ANY LINUX MACHINE)

The systemd service unit ensures the CCTV dashboard runs 24/7 in the background, recovers automatically from any unexpected crash, and boots automatically upon server power-on.

> **One-Click Automatic Systemd Setup (Recommended)**
> You can install and start the systemctl service immediately with one command from the project root:
> `sudo ./install-systemd.sh`
> *To uninstall later, run: `sudo ./uninstall-systemd.sh`*

5.1: The Production Service Unit File (`/etc/systemd/system/inncctv.service`)

```
[Unit]
Description=Innspark Enterprise 24-Camera Surveillance Operations Center
After=network.target remote-fs.target nss-lookup.target

[Service]
Type=simple
User=inncctv
Group=inncctv
WorkingDirectory=/opt/inncctv
ExecStart=/usr/bin/env node /opt/inncctv/server/index.js
Restart=always
RestartSec=5s

# Production Environment Variables
Environment=NODE_ENV=production
Environment=PORT=3000
Environment=ADMIN_PASSWORD=innspark@2026
Environment=INNCCTV_SECRET=innspark_secure_production_token_secret_key_2026

# CIS Benchmark Linux Sandboxing
ProtectSystem=full
ProtectHome=true
NoNewPrivileges=true
PrivateTmp=true
LimitNOFILE=65535

[Install]
WantedBy=multi-user.target
```

5.2: Complete Systemctl Command Reference

Action	Linux Command	Description
Install Unit File	`sudo cp /opt/inncctv/deploy/inncctv.service /etc/systemd/system/`	Copies unit file to systemd directory
Reload Systemd	`sudo systemctl daemon-reload`	Notifies systemd of new/changed service unit
Enable Auto-Boot	`sudo systemctl enable inncctv.service`	Configures service to launch on server boot
Start Service	`sudo systemctl start inncctv.service`	Launches dashboard & streaming gateway
Check Live Status	`sudo systemctl status inncctv.service`	Displays active state, uptime, PID & recent logs
Restart Service	`sudo systemctl restart inncctv.service`	Gracefully restarts dashboard & camera streams
Stop Service	`sudo systemctl stop inncctv.service`	Safely stops all monitoring & camera feeds

Action	Linux Command	Description
Check Boot Status	`sudo systemctl is-enabled inncctv.service`	Verifies service will start on server reboot
Stream Live Logs	`sudo journalctl -u inncctv.service -f`	Follows live console & camera connection logs
Inspect Errors	`sudo journalctl -u inncctv.service -p err -e`	Filters and displays recent error events

Universal Compatibility Note

By utilizing `ExecStart=/usr/bin/env node`, the service automatically detects the correct Node.js binary location on any Linux distribution (whether installed under `/usr/bin/node`, `/usr/local/bin/node`, or custom paths).

6. NGINX REVERSE PROXY & SSL/TLS CONFIGURATION

To serve the dashboard over standard port 80/443 with WebSocket upgrading and TLS encryption:

```
# Copy pre-configured Nginx reverse proxy configuration
sudo cp /opt/inncctv/deploy/nginx-inncctv.conf /etc/nginx/sites-available/inncctv
sudo ln -sf /etc/nginx/sites-available/inncctv /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t && sudo systemctl restart nginx

# Obtain Free SSL Certificate via Let's Encrypt / Certbot
sudo apt-get install -y certbot python3-certbot-nginx
sudo certbot --nginx -d cctv.yourcompany.com
```

7. VAPT & SECURITY HARDENING VERIFICATION

VAPT Compliance Summary

The dashboard includes active defenses for OWASP Top 10 web vulnerabilities including anti-brute-force rate limiting, scrypt salted password hashing, HMAC-SHA256 tokens, strict CSP headers, and input sanitization against shell command injection.

Verification Commands (Run in Terminal to Validate):

```
# 1. Verify Security Headers (Information Disclosure Protection)
curl -s -I http://localhost:3000/ | grep -E "(X-Frame|X-Content|X-XSS|Content-Security|X-Powered-By)"

# 2. Test Input Injection Sanitization (Should return HTTP 400 Bad Request)
curl -s -X POST http://localhost:3000/api/cameras \
  -H "Content-Type: application/json" \
  -d '{"name":"Test","floor":1,"ip":"invalid-ip","rtspUrl":"rtsp://evil;rm -rf /"}'

# 3. Test Anti-Brute-Force Rate Limiting (Triggers lockout after 5 failed attempts)
for i in {1..6}; do
  curl -s -X POST http://localhost:3000/api/auth/login \
    -H "Content-Type: application/json" \
    -d '{"username":"admin","password":"wrongpassword"}'
  echo ""
done
```

8. OPERATIONS, BACKUP & CREDENTIALS

Default Access: URL: `http://<SERVER_IP>:3000` | User: `admin` | Password: `innspark@2026`

```
# Backup 24-camera configuration to file
sudo cp /opt/inncctv/config/cameras.json /opt/cameras_backup_$(date +%F).json

# Restore configuration from backup
sudo cp /opt/cameras_backup_2026-10-15.json /opt/inncctv/config/cameras.json
sudo systemctl restart inncctv.service
```